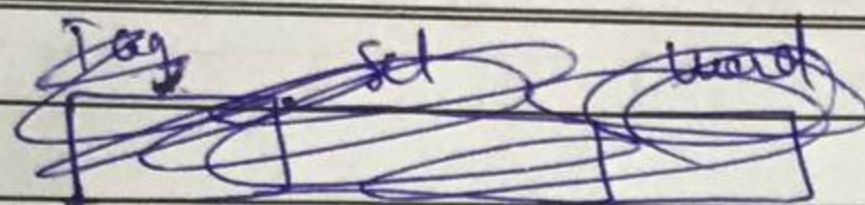




Course Computer Organization & Architecture Answer Sheet No. 28061
Student's Name M. Yasar Signature [Signature]
Roll No. 152-4507 Section EE-1 Date 15th Nov 2018

Q.1 a,



Cache, lines per set = 2
line size = 16 bytes
total size = 128K bytes

$$\text{total lines} = \frac{128K}{16} = 8192 = \underline{8K \text{ lines}}$$

$$\text{no. of sets} = \frac{8K}{2} = 4K \text{ sets}$$

~~no. of set bits = log~~

$$\text{no. of set bits} = \log_2(4 \times 1024)$$

$$\boxed{= 12}$$

$$\text{line size} = \text{block size} = 16 \text{ bytes}$$

$$16 = 2^4 \Rightarrow \boxed{4 \text{ bits per word}}$$

main
total memory = 64M

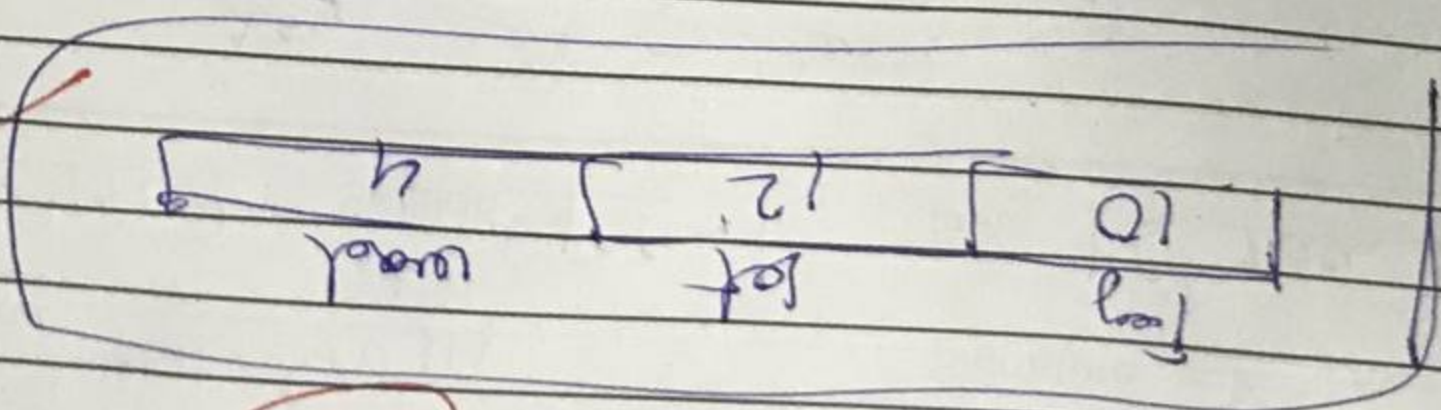
$$\text{no. of blocks} = \frac{64M}{16} = 4M$$

$$\text{no. of bits required for block} = \log_2(4M)$$

$$= 22$$

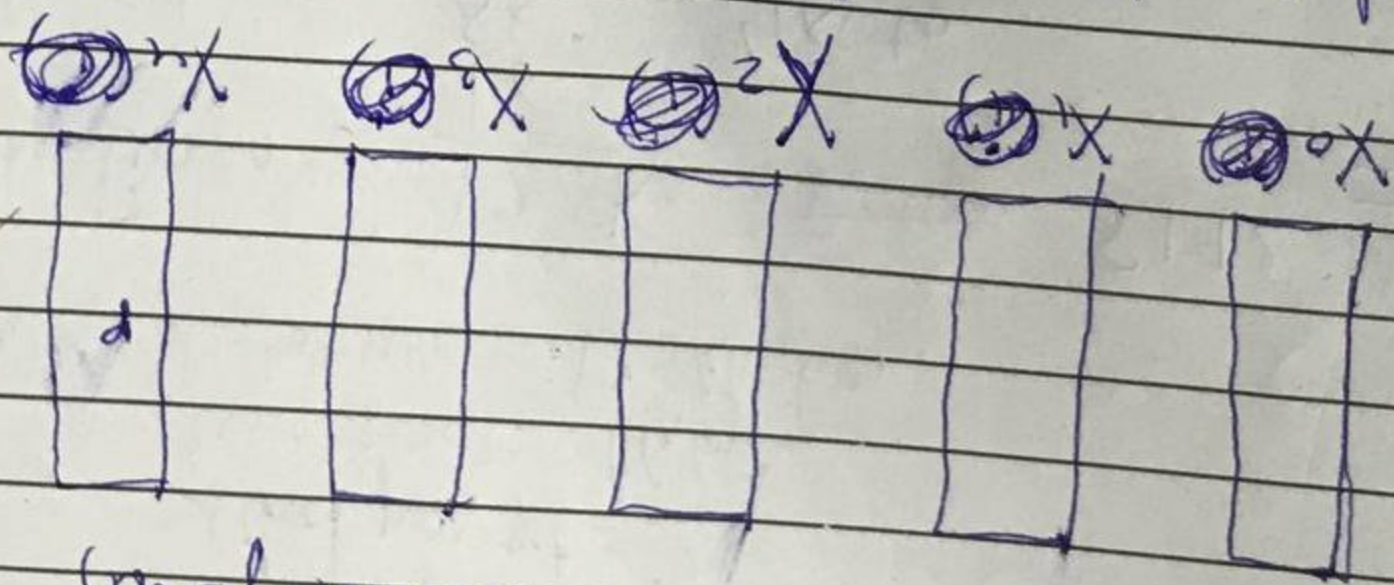
$$\boxed{\text{no. of tag bits} = 22 - 12 = 10}$$

hence $\text{tag} = 10$
 $\text{set} = 12$
 $\text{word} = 4$



Qb/

RAID 3 $\Rightarrow$ bit level parity



where $i = \text{no of bit}$

parity can help recovering of corrupted bit

$$X_4(i) = X_0(i) \oplus X_1(i) \oplus X_2(i) \oplus X_3(i)$$

where $\oplus = \text{exclusive-OR}$

6

$\rightarrow$ add $X_4(i)$ on both sides

$$X_4(i) \oplus X_4(i) = X_0(i) \oplus X_1(i) \oplus X_2(i) \oplus X_3(i) \oplus X_4(i)$$

$$\rightarrow \text{LHS} = X_4(i) \oplus X_4(i) = 0$$

$\rightarrow$ add XOR with $X_4(i)$ on LHS & RHS

$$X_2(i) = X_0(i) \oplus X_1(i) \oplus X_2(i) \oplus X_3(i) \oplus X_4(i) \oplus X_2(i)$$

$$\rightarrow X_2(i) \oplus X_2(i) = 0$$

hence we get

Q / Part No.

Rough
Work

$$X_2(i) = X_0(i) \oplus X_1(i) \oplus X_3(i) \oplus X_4(i)$$

The above equation can help to
recover the corrupted data of X_2

Q.2 $M = \text{Multiplier} = -9 = 1011$ ($n=5$)
 $a = \text{Multiplicand} = 14 = 0110$

$$(+9)_{10} = (01001)_2 \quad (+14)_{10} = (01110)_2$$

$$(-9)_{10} = (10111)_2 \quad (-14)_{10} = (10010)_2$$

00 $\rightarrow$ ASR
 01 $\rightarrow A \leftarrow A + M$
 10 $\rightarrow A \leftarrow A - M$
 11 $\rightarrow$ ASR

n	A	Q	Q _{n-1}	Restripts	
5	00000	01110	0	ASR	00000
4	00000	00111	0	ASR	01001
	01001	00111	0	$A \leftarrow A + M$	01001
3	00100	10011	1	ASR	00001
2	00010	01001	1	ASR	10111
1	00001	00100	1	$A \leftarrow A + M$	11000
	11000	00100			10111
					11110
					10101

n	A	Q	Q _{n-1}	Restripts	
5	00000	01110	0	ASR	
4	00000	00111	0	$A \leftarrow A - M$	
	10111	00111	0	ASR	
3	11011	10011	1	ASR	
2	11101	11001	1	$A \leftarrow A + M$	
1	11110	11100	1		
	10101	11100	1		
	11010	11110	0		

n	A	Q	Q-1	Description
5	00000	01110	0	ASR
4	00000	00111	0	ASR
3	00000	00011	1	ASR

n	A	Q	Q-1	Description
5	00000	01110	0	ASR
4	00000	00111	0	$A \leftarrow A - M$
3	01001	00111	0	ASR
2	00100	10011	1	ASR
1	00010	01001	1	ASR
0	00001	00100	1	$A \leftarrow A + M$
	11000	00100	1	
0	11100	00010	0	ASR

$$111100 \ 00010 \Rightarrow = 126$$

2's comp

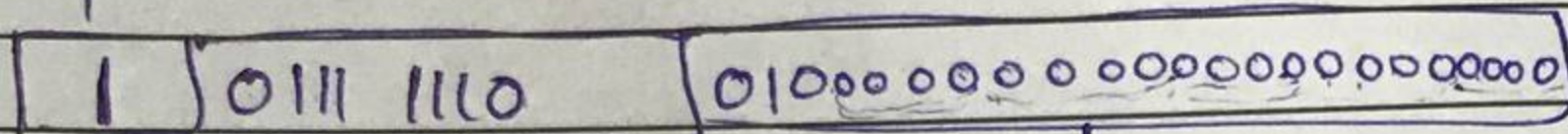
$$(00011 \ 11110)_2 = (126)_{10}$$

Q. 5a) -0.625

$$(0.625)_{10} = (0.101)_2$$

$$-0.101 \Rightarrow \boxed{1.01000 \times 2^{-1}}$$

normalized form

sign bit
78 bit
exponent (E')

23 bit significand

 $\ominus 0.625$ hence sign bit = 1

$$\text{exponent} = -1$$

$$E' = \text{exponent} + 127 = -1 + 127 = 126$$

$$\boxed{(126)_{10} = (01111110)_2}$$

Writing in hexa (making groups of four)

~~01111110~~

11011	1111	0010	0000	0000	0000	0000	0000
-------	------	------	------	------	------	------	------

B	F	2	0	0	0	0	0
---	---	---	---	---	---	---	---

 $\Rightarrow$ Hexa
6

(16)

(-16)

(90)

(-90)

$$(165)_{10} = (010100101)_2$$

$$(-165)_{10} = (101011011)_2$$

$$(90)_{10} = (001011010)_2$$

$$(-90)_{10} = (110100110)_2$$

adding -165 & -90

~~0000~~

$$\begin{array}{r} 101011011 \\ + 110100110 \\ \hline 100000001 \end{array}$$

1

Carry (overflow)

$$(100000001)_2$$

take two's complement

$$(01111111)_2 = (255)_{10}$$

$$\text{as } (-165) + (-90) = -(255)$$

Hence correct ans

overflow detected as carry out is present.

National University of Computer and Emerging Sciences, Lahore Campus



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Program:	BS (Electrical Engineering)	Semester:	Fall 2018
Duration:	60 Minutes	Total Marks:	30
Paper Date:	15-Nov-2018	Weight	15%
Section:	EE	Page(s):	1
Exam:	Midterm-2	Roll No:	

Instruction/Notes: Solve All Questions

Q.1a A two-way set-associative cache has lines of 16 bytes and a total size of 128 Kbytes. The 64-Mbyte main memory is byte addressable. Show the format of main memory addresses. [CLO 2][5]

Q.1b In RAID level 3 consider an array of five disk drives in which drive X0 through X3 contain data and drive X4 is the parity disk. If drive X2 fails, show how data will be reconstructed from the remaining drives. *bit wise parity* [CLO 2][5]

Q.2a Use the Booth algorithm to multiply -9 (multiplicand) by 14 (multiplier), where each number is represented using 5-bits. [CLO 3][5]

Q.2b Define overflow condition. Is the following a valid alternative definition of overflow in twos complement arithmetic? Give reasons. [CLO 3][5]

"If the exclusive-OR of the carry bits into and out of the leftmost column is 1, then there is an overflow condition. Otherwise, there is not."

Q.3a Express the following number in IEEE 32-bit floating-point format. Give your answer in hexadecimal. [CLO 3][5]

- 0.625

Q.3b Assume the numbers are represented in 9-bit twos complement representation. Show the calculation of the following: [CLO 3][5]

$$(-165) + (-90)$$

Determine if there is an overflow.

$$\frac{2^{11} - 1}{2^n} = \dots$$

13
2

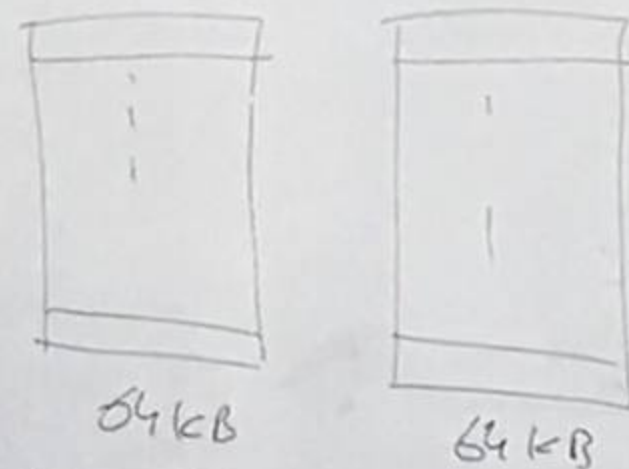
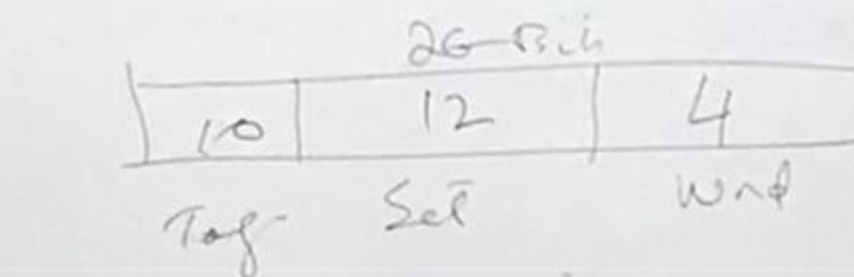


COA FALL 2018

MID 2 QUESTION 1a.

$L = 16 \text{ Bytes}$
 Total Size = 128 KBytes.
 M.M = 64 MB

$$2^{26} = 64 \text{ M}$$



$$\begin{aligned}
 \text{No. of Set } \frac{64 \text{ kB}}{16 \text{ B}} \\
 = \frac{2^{16}}{2^4} = 2^{12}
 \end{aligned}$$

$$X_0(1) \oplus X_1(1)$$

CDA mid #02

For n bit number :

$$\frac{1}{2^{n-1}} \left(2^{n-1} - 1 \right)$$

Q#02(a) Booth's Algorithm

$-M = 9 \rightarrow 01001$
 $M = -9 = 10111$
 $Q = 14$

A	Q	Q _{n-1}	Steps
00000	01110	0	0000
00000	00111	0	0000 ASR
01001	00111	0	A ← A - M
00100	10011	1	ASR
00010	01001	1	ASR
00001	00100	1	ASR
11000	00100	1	A ← A + M
11100	00010	0	ASR

Q#02(b) overflow conditions:

Definition (2 positive/negative ...)

If carry is 1 (ans is negative)
 If carry is 0 (ans is +ve)

0xxxxx

1xxxxx

case 1:

2 ~~generate~~ signs k
 case mn MSB can't
 generate a carry

Date: _____

case 2^a:

In case of 2 negative numbers:

$$\begin{array}{r} 1 \quad 1 \\ \swarrow \quad \nwarrow \\ 1 \quad x \quad x \quad x \quad \dots \quad x \\ 1 \quad x \quad x \quad x \quad \dots \quad x \\ \hline 0 \end{array} +$$

case 2^b:

$$\begin{array}{r} 1 \quad 1 \\ \swarrow \quad \nwarrow \\ 1 \quad x \quad x \quad \dots \quad x \\ 1 \quad x \quad x \quad \dots \quad x \\ \hline 1 \quad x \quad x \quad \dots \quad x \end{array}$$